Excluding endpoint touching bands

Human Compression project: L044 continuation

Use the standing hypotheses of the accepted nested-bridge theorem: $G \le A_n$, $n \ge 12$, is transitive with a connected string C-marking and all maximal omissions intransitive. Suppose its high-rank equality $g = h$ occurs, with one crossing at $i = 0$ (the other orientation is the literal reversal). The unique edges are $a_0 -_0 b_0$ and $c -_j d$, cutting the full graph into A, M, D. Thus

$$n = 2r, \quad |A| = h + 1, \quad |M| = j, \quad |D| = 2r - j - h - 1, \quad 0 < h < j.$$

The right suffix $G_{>j}$ fixes $A \cup M$. Retain the exact last/first shared-colour hypotheses: h is last shared across the 0-cut, and $g = h$ first shared across the j-cut. Consequently

$$\rho_0|_D = 1, \qquad \rho_j|_A = 1, \qquad G_{>h} \text{ fixes } A. \tag{1}$$

All claims below keep these genuine purity conditions.

Lemma 1 (The endpoint forces a three-point outside orbit)**.** *Under these assumptions $h = 2$, and, writing the forced ordered paths as below,*

$$\rho_0 = (a_0\ b_0)(a_1\ a_2). \tag{2}$$

Proof. The accepted fixed-prefix path in A, including its last h-edge, has at least $h + 1$ vertices and therefore exhausts A. The middle path between the bridges similarly exhausts M. Each intermediate colour occurs exactly once, and their order is forced from the first anchor: all colours at least two fix that anchor; after following colour one all colours at least three fix the next point, by commutation; continuing gives

$$A:\ a_0 -_1 a_1 -_2 \cdots -_h a_h, \qquad M:\ b_0 -_1 b_1 -_2 \cdots -_{j-1} b_{j-1} = c.$$

The suffix of the middle path from b_1 to c, followed by the j-edge to d, uses only indices at least two. The element ρ_0 fixes d by (1) and commutes with every generator on that path. Hence it fixes $M \setminus \{b_0\}$ pointwise, as well as all of D. Its unique cross-orbit transposition is $(a_0\ b_0)$, so its remaining action is inside $A' = \{a_1, \ldots, a_h\}$.

For $h = 1$ that remaining set is a singleton, making ρ_0 odd. For $h \ge 3$, the group $\langle \rho_2, \ldots, \rho_h \rangle$ is transitive on A' by the displayed path. Its subgroup $\langle \rho_3, \ldots, \rho_h \rangle$ has exactly one fixed point in A', namely a_1: all other points are moved by displayed edges of those colours, while a_1 is fixed by commutation from the anchor. The restriction of ρ_0 centralizes both groups, so fixes a_1 and then all of A'. This again makes it odd. Thus $h = 2$, and evenness forces (2). □

Theorem 2 (No endpoint touching equality)**.** *The high-rank equality $g = h$ is impossible when at least one crossing is an endpoint and the stated first/last shared-colour and orientation hypotheses hold.*

Proof. If $j = r - 1$, apply the lemma in both index directions. It gives $h = 2$ and $j - h = 2$, so $r = 5$ and $n = 10$, contrary to $n \ge 12$.

Suppose $j < r - 1$. Strictness in the reversed outside-orbit bound forces $h = j - 1$. Thus $h = 2$, $j = 3$. Put $l = r - j = r - 3 \ge 2$. Then $|D| = 2l$, $M = \{b_0, b_1, c\}$, and on $A \cup M$ the first three generators are exactly

$$\rho_0 = (a_0\ b_0)(a_1\ a_2), \quad \rho_1 = (a_0\ a_1)(b_0\ b_1), \quad \rho_2 = (a_1\ a_2)(b_1\ c)\delta, \tag{3}$$

where $\delta = \rho_2|_D$. Indeed ρ_1 fixes D because $g = 2$ is the first shared colour, and it fixes c by the unique 3-crossing; the remaining actions in (3) follow from the three-point paths. Write $R = G_{>3}$. The group $\langle \delta, R \rangle$ is transitive on D, and δ centralizes R. The same critical commuting-orbit argument as in the accepted interior note applies on this side alone: δ is fixed-point-free, exchanging l pairs, and either R is transitive on D or it has two faithful natural S_l orbits of size l. To spell out the count, a transitive R has essential independent pair images of length $l - 1$, hence full S_l image by Whiston; an intransitive R has faithful orbits of size at least l, hence exactly two, again S_l. This uses only the actual global omission argument and not an intersection property of a projection.

Equation (3) and evenness make l even. Since $n = 2l + 6 \geq 12$, it follows that $l \geq 4$. Set $t = \rho_3$ and $Q = \langle t, R \rangle = G_{>2}$. The element t fixes A by (1), fixes b_0 by the unique 0-crossing, and fixes $b_1 = b_0^{\rho_1}$ by commutation with ρ_1. Hence $t = (c\ d)v$, where v is supported on $D \setminus \{d\}$. We next show that Q is transitive on $Z = D \cup \{c\}$. If R is transitive on D, the cd-edge proves this immediately. Otherwise identify the two S_l rows through δ. The subgroup $H = G_{>4}$ fixes $d, d' = d^\delta$. Its pair images form an independent family of length $l - 2$ in S_{l-1}, so Whiston's equality theorem makes its action on the remaining $l - 1$ pairs the full S_{l-1}. In this two-row case it is the same natural action in both rows. The involution v centralizes H and fixes d': it preserves $D \setminus \{d\}$, whose only H-fixed point is d'. On the two remaining rows of length $l - 1 \geq 3$, a commuting permutation is either the identity or the exchange identifying those rows, because the natural S_{l-1} centralizer is trivial. Since t is even, v must be odd and therefore is that row exchange. Thus t joins the two R-orbits as well as adjoining c, proving transitivity.

The actual supported C-group $Q \leq A_Z$ has degree $2l + 1$ and rank l, with the anchor c fixed by R. Every maximal omission inside Q is intransitive on Z: otherwise the corresponding global omission is already transitive on Z, and adjoining its discarded generator, supported on Z, cannot join omission orbits, contrary to transitivity of G. If $|Z| = 9$, the complete accepted small supported-tail lemma already excludes Q: at this rank its only small survivors have degrees five and six. That lemma's proof uses the actual extension anchor, the complete small primitive attachment certificate and the small anchored-imprimitive bound; it uses no smaller-degree alternating induction. Thus we may assume $|Z| \geq 13$. The accepted proper-primitive bound and anchored-imprimitive bound then both exclude a proper transitive $Q < A_Z$ at this rank. Consequently

$$Q = A_Z. \tag{4}$$

All required small census programs and classical premises of those exact predecessors remain paid; no new census is invoked.

The six-point core in (3) is exactly the accepted Table 2(2) A_5 marking. Let $P = \langle \rho_0, \rho_1, \rho_2 \rangle$. It is a subgroup of $A_5 \times \langle \delta \rangle$, surjecting onto both factors. If $\delta \notin P$, it would be the graph of a nontrivial homomorphism $A_5 \to C_2$, impossible by perfection of A_5. Thus $\delta \in P$. But δ is even, nontrivial and supported on D, so (4) puts it in Q too. Full IP gives $P \cap Q = 1$ because their distinguished index sets are disjoint. This is the final contradiction. Reversing handles a right endpoint crossing. □

Attribution and exact boundary. This is a new replacement argument for the endpoint touching case left by the two-bridge count in Cameron–Fernandes–Leemans–Mixer, *Highest rank of a polytope for A_n*, arXiv:1605.09173v1, Proposition 5.28, physical page 37. All accepted path, outside-orbit, critical-pair, proper-primitive, anchored and small-tail proofs are charged as whole dependencies, including their finite programs and retained classical inputs. The proof does not reclassify the source tables or repeat those computations.

Together with the accepted interior touching argument, it forces $g > h$ in the oriented high-rank bridge situation. The subsequent $g > h$ support and middle-orbit equality analysis remains separate. There is no whole L044, new formal theorem, original, question or priority credit in this bounded packet.